MIL-B-2766C
23 March 1987
SUPERSEDING
MIL-B-2766B
22 August 1967
(See 6.6)

# MILITARY SPECIFICATION

# BATTING, FIBROUS GLASS, LIFESAVING EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers fibrous glass batting for use in life preservers.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| UU-P-268 | - Paper, Kraft, Wrapping. |
| VV-D-1078 | - Damping Fluid, Silicone Base (Dimethyl Polysiloxane). |
| CCC-C-446 | - Cloth, Muslin, Cotton. |
| PPP-B-566 | - Boxes, Folding, Paperboard. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-B-640 | - Boxes, Fiberboard, Corrugated, Triple-Wall. |
| PPP-B-665 | - Boxes: Paperboard, Metal Edged and Components. |
| PPP-B-676 | - Boxes, Setup. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-C-3955 | - Cans, Composite, Spirally Wound. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 4220
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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to this extent specified. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1448 - Standard Test Method for Micronaire Reading of Cotton Fibers.
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material. The batting shall consist of fibrous glass and a binder. A silicone base water repellent shall also be added. The batting shall be uniform in appearance and quality, and shall not contain materials which may cause injury or damage. Asbestos and components containing asbestos shall be prohibited. When specified in the contract or order, a certificate of compliance shall be prepared to show that the batting is asbestos-free (see 6.2.2).

3.3 Fiber diameter. The average diameter of 50 randomly selected fibers shall be 0.00008 ± 0.00002 inch (see 4.5.1).



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3.4 Weight. Batting shall be furnished by weight (see 6.2.1 and 4.5.2).

3.5 Water repellent. Each lot shall be made water repellent in accordance with VV-D-1078. The water prepellent of each lot shall be not less than 1 percent by weight of the batting. When specified in the contract or order, a certificate of compliance shall be prepared (see 4.5.3 and 6.2.2).

3.6 Binder. For each lot the binder shall be between 4 and 12 percent weight of the batting (see 4.5.4). When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

3.7 Ignition loss. Ignition loss shall be between 5 and 15 percent by weight of the batting (see 4.5.4).

3.8 Buoyancy. Buoyancy of the test pad shall be at least 4.8 pounds (see 4.5.5).

3.9 Workmanship. The batting shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability or durability (see 4.4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified herein.



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4.3 First article inspection. First article inspection shall consist of the examination of 4.4.3 and tests specified in 4.5 and table I (see 6.3).

TABLE I. First article inspection.

| Inspections | Requirement paragraph | Test paragraph |
|---|---|---|
| Fiber diameter | 3.3 | 4.5.1 |
| Weight | 3.4 | 4.5.2 |
| Water repellent | 3.5 | 4.5.3 |
| Binder | 3.6 | 4.5.4 |
| Ignition loss | 3.7 | 4.5.4 |
| Buoyancy | 3.8 | 4.5.5 |
| Workmanship | 3.9 | 4.4.3 |

4.3.1 First article test report. When specified in the contract or order, a first article test report shall be prepared (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.2 and table II.

TABLE II. Quality conformance inspection.

| Inspection | Requirement paragraph | Test paragraph | Unit of product |
|---|---|---|---|
| Group A | | | |
| Fiber diameter | 3.3 | 4.5.1 | Lot |
| Weight | 3.4 | 4.5.2 | Lot |
| Workmanship | 3.9 | 4.4.3 | Lot |
| Group B | | | |
| Water repellent | 3.5 | 4.5.3 | Lot |
| Buoyancy | 3.8 | 4.5.5 | 2,000 pounds |
| Group C | | | |
| Ignition loss | 3.7 | 4.5.4 | Lot |

4.4.1 Lot. A lot shall consist of all batting produced in a continuous run or at the same time under the same conditions and offered for delivery under one contract.

4.4.2 Sampling. Sampling shall be as specified in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Sampling for quality conformance tests. For quality conformance inspection, a random sample shall be selected from each lot in accordance with MIL-STD-105, level II, with an acceptable quality level (AQL) of 6.5 percent defective. To determine compliance with the requirements as specified in 3.2 and 3.8, the Government reserves the right to select for the buoyancy test (see 4.5.5) any questionable area from each 2,000-pound unit of product at inspection level II with an AQL of 2.5 percent.



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4.4.2.2 Rejected lots. If a sample fails to pass group B and C inspections, the contractor shall notify the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. After the corrective action has been taken, group B and C inspections shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until group B and C inspections have shown that the corrective action was successful.

4.4.3 Examination for visible defects. An examination for visible defects shall be performed in order to verify conformance to the requirements as specified in 3.9.

4.4.3.1 Sampling for examination for packaging. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-STD-105. The AQL shall be 2.5 percent defective (see table III).

TABLE III. Examination of packaging defects.

| Examine | Defects |
|---|---|
| Packaging | Not the level specified; not in accordance with contract requirements. |
| | Not packaged in unit containers as specified. |
| | Mixed size in the same container; containing less than indicated of specified quantity. |
| Packing | Not the level specified; not in accordance with the contract requirements. |
| | Packing material not as specified; closures not accomplished by specified methods or materials. |
| | Gross weight of shipping container exceeds specified limit. |
| | Any nonconforming components, component missing, damaged, or otherwise defective affecting serviceability. |
| Marking | Interior and exterior marking omitted, illegible, incorrect, incomplete, not in accordance with contract requirements. |



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4.5 Test procedures. Test procedures shall be as specified in 4.5.1 through 4.5.5.

4.5.1 Fiber diameter. The diameter of the fiber shall be determined by either of the following methods (see 3.3):

(a) Microscopic. Diameter of fibers shall be determined microscopically. Fifty random fibers shall be measured. The average diameter shall be the average of all measurements.
(b) Air flow. The air flow method as measured by the micronaire instrument in accordance with ASTM D 1448, with the addition that the micronaire unit shall be calibrated for the purpose of testing fibrous glass.

4.5.2 Weight. Batting shall be weighed by using a standard scale calibrated to 0.01 pound (0.5 milligrams (mg)) (see 3.4).

4.5.3 Water repellent. Water repellent shall be as specified in 4.5.3.1 and 4.5.3.2.

4.5.3.1 Extraction. A 10-gram sample of batting (weighed to the nearest mg) shall be cut to fit into an extraction siphon cup lined with a folded filter paper. Using a solvent extraction apparatus, 60 to 70 milliliters (mL) of petroleum ether shall be extracted for 4 hours at such a rate that approximately 3 minutes shall be required to fill and empty the extraction cup. The extracted sample shall be saved for the ignition loss test. The petroleum ether shall be transferred to a tared beaker. While passing a gentle current of filtered air over the beaker to prevent boiling, the petroleum ether shall be evaporated to dryness on a steam bath. It shall be dried at 100 ± 5 degrees Celsius (°C) for 1 hour. The extract shall be saved for infrared spectro-identification (see 4.5.3.2).

4.5.3.2 Infrared identification of extract (test for silicone water repellent). An infrared spectrum of the petroleum ether extract shall be obtained on a double-beam recording infrared spectrophotometer operating from 2 to 16 micrometers (µm). Cell thickness shall be selected so that the absorption at 7.9 µm lies between 0.8 and 1.0. The major characteristic absorption bands shall be only at 3.4, 7.9, 9.2, 9.8, 11.5, and 12.5 µm (see 3.5).

4.5.4 Ignition loss. The sample of fibrous glass from which the silicone has been extracted (see 4.5.3.1) shall be dried in a circulating air oven at 100°C for 30 minutes, cooled, and weighed. It shall be placed in a suitable container and shall be ignited to constant weight at 537°C (1,000 degrees Fahrenheit (°F)) (see 3.6 and 3.7).

$$\text{Ignition loss, percent} = \frac{\text{Initial weight (before extraction)} - \text{Final weight (after ignition)}}{\text{Initial weight}} \times 100$$



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4.5.5 Buoyancy. Ten ounces of fibrous glass shall be stuffed into a rectangular bag made in accordance with CCC-C-446 type I, class 1 cotton muslin cloth enclosing an area 9-1/2 inches wide and 12 inches long, when flat. After securing the bag within a suitably weighted wire frame, it shall be submerged vertically in fresh water at 21 ± 5.6°C (70 ± 10°F) for 24 hours. The top of the bag shall be kept 2 to 3 inches below the surface. The buoyancy shall be expressed in pounds, as the apparent weight of the weighted empty frame in water minus the apparent weight in water of the assembly of the glass-filled bag and the weighted frame, after the 24-hour submersion (see 3.8).

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2.1).

5.1.1 Level A. Fibrous glass batts shall be rolled and individually wrapped in 40-pound basis weight Kraft paper (24 times 36 minus 500) in accordance with grade A of UU-P-268 and sealed with a suitable Kraft gummed tape or adhesive.

5.1.1.1 Unit containers. Except for unit containers in accordance with MIL-P-116 for the applicable method of preservation, fibrous glass batts (after wrapping) shall be packaged in one of the unit containers in accordance with methods shown in PPP-B-566, PPP-B-636, PPP-B-665, PPP-B-676, or MIL-C-3955. Container closure and sealing shall be in accordance with the applicable container specification or appendix thereto.

5.1.1.2 Intermediate packaging. Unit quantities in an intermediate container shall be as specified (see 6.2.1). Intermediate containers shall be in accordance with any of the following specifications at the option of the contractor: PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676. Container closure and sealing shall be in accordance with the applicable container specification or appendix thereto. The gross weight of paperboard boxes shall not exceed 10 pounds; fiberboard containers shall not exceed 20 pounds.

5.1.2 Commercial. Fibrous glass batts shall be rolled, individually wrapped and packaged in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level B or C, as specified (see 6.2.1).

5.2.1 Level B. Fibrous glass batts packaged as specified (see 6.2.1), shall be packaged in containers in accordance with PPP-B-636, class weather-resistant, or PPP-B-640, class 2, at the option of the contractor. Caseliners for fiberboard boxes may be omitted provided all center and edge seams and manufacturer's joints are sealed and waterproofed with pressure-sensitive tape



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in accordance with the applicable fiberboard box specification. Shipping containers shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. Containers, in accordance with PPP-B-636, shall not exceed the weight limitations of the specification. When specified, intermediate fiberboard containers in accordance with the weather-resistant class of PPP-B-636, closed, sealed and banded as specified herein, shall be used as the shipping container and shall not be overpacked.

5.2.2 Level C. Fibrous glass batts, packaged as specified (see 6.2.1), shall be packed in containers in accordance with PPP-B-636, class domestic, or PPP-B-640, class 1, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of the fiberboard boxes shall not exceed the weight limitations of the applicable box specification; intermediate fiberboard containers in accordance with PPP-B-636, closed, sealed and banded, as specified herein, shall be used as the shipping container and shall not be overpacked.

5.2.3 Commercial. Fibrous glass batts shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 for levels A, B, and C, in accordance with ASTM D 3951 for commercial.

6. NOTES

6.1 Intended use. The fibrous glass batts covered by this specification are intended to be used as the buoyant filler material in life preservers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article sample, when required (see 3.1).
(c) Total weight of the batting required (see 3.4).
(d) Test conditions (see 4.2.1).
(e) Level of preservation and packing required (see 5.1 and 5.2).
(f) Special marking, when required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.



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| Paragraph no. | Data requirement title | Applicable DID no. | Option |
|---|---|---|---|
| 3.2, 3.5, 3.6 | Certificate of compliance | DI-E-2121 | --- |
| 4.3.1 | First article inspection report | DI-T-4902 | --- |

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Batting
Fibrous glass
Life preserver

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

User activity:
Navy - CG

Preparing activity:
Navy - SH
(Project 4220-0312)



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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-B-2766C | Batting, Fibrous Glass, Lifesaving Equipment |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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